

Deep Reinforcement Learning for Thai Stock Portfolio Allocation

Final Pilot Methodology and Results

Final Project Plan

May 31, 2026

1 Objective

This project implements a reproducible pilot pipeline for deep reinforcement learning portfolio allocation in Thai stocks. The current implementation validates the end-to-end experimental workflow on a five-stock Thai OHLCV starter universe and on a deterministic synthetic pilot universe. Both use the same trading environment, metrics, baseline policies, and chronological train/validation/test split logic.

The reported results are pilot results. They validate the research pipeline, but they should not be interpreted as a complete Thai-market performance claim because only starter SET index, sector-context, daily macro-proxy, and Yahoo statement fundamental features are integrated with overlapping history. Sentiment and official macro ingestion/merge scaffolding are implemented, but the available Yahoo latest-news probe and BOT web-table probe do not overlap the 2021–2024 modeling window. Historical official macro releases, licensed fundamentals, historical sentiment, and a broader Thai universe remain incomplete.

2 Data

The real-data starter run uses Yahoo Finance OHLCV data for ADVANC.BK, AOT.BK, CPALL.BK, KBANK.BK, and PTT.BK. The real feature table is:

```
/lustrefs/project/25sfcs03/drl_thai_stock/data/processed/features_real_ohlc  
.parquet
```

It contains 4,840 rows, five tickers, and 21 columns from 2021-01-04 to 2024-12-30. The latest verified real comparison used the split in Table 1.

Table 1: Real OHLCV chronological split.

Split	Date range	Rows	Unique dates	Tickers
Train	2021-01-04 to 2023-05-29	2,900	580	5
Validation	2023-05-30 to 2024-03-12	970	194	5
Test	2024-03-13 to 2024-12-30	970	194	5

The generated data-quality report shows 968 unique trading dates per ticker, zero missing panel dates for each ticker, and zero missing values across the active feature columns. It is stored at:

```
/lustrefs/project/25sfcs03/drl_thai_stock/reports/data_quality/real_ohlc  
v_data_quality.md
```

3 Features

The active technical feature set includes one-day, five-day, and twenty-day returns; twenty-day volatility; ten-day and thirty-day moving-average ratios; RSI; MACD; Bollinger z-score; volume ratio; and ATR. Expanded pilot feature tables add SET index features, sector-context features, daily macro-proxy features, and lagged Yahoo statement fundamentals. Sentiment and official macro feature tables are also buildable through no-lookahead merge scaffolds, but current Yahoo latest-news and BOT web-table rows are outside the historical validation window.

4 Environment

The trading environment is a Gymnasium-compatible long-only portfolio allocator. Actions are continuous portfolio weights over the stock universe. The environment applies transaction costs and tracks cash, portfolio value, net return, turnover, drawdown, and per-ticker weights.

The reward combines portfolio return with turnover and drawdown penalties:

$$r_t = w_r R_t - \lambda_{\text{turnover}} T_t - \lambda_{\text{drawdown}} D_t$$

The current real-data configs use return weight $w_r = 1.0$, turnover penalty $\lambda_{\text{turnover}} = 0.05$, and drawdown penalty $\lambda_{\text{drawdown}} = 0.1$.

5 Models and Baselines

The implemented DRL models are PPO and A2C, with support paths for DDPG, TD3, and SAC. The implemented baselines are equal weight, random allocation, momentum, per-ticker buy-and-hold, moving-average crossover, and mean-variance allocation.

Hyperparameter search uses SLURM job arrays and the trial runner in `src/experiments/run_trial.py`. The search space covers learning rate, gamma, GAE lambda, entropy coefficient, rollout length, PPO clip range, and PPO batch size.

6 Verified Runs

Table 2: Verified and invalidated HPC runs.

Purpose	HPC path	Status
Synthetic split-aware comparison	<code>results/algo_compare_2853</code>	Valid
Real OHLCV PPO/A2C comparison	<code>results/real_ohlcvc_compare_2855</code>	Valid
Optuna sanity array	<code>results/optuna_sanity_2856</code>	Valid
Fixed Optuna search	<code>results/optuna_search_fixed_2863</code>	Valid
Longer best-parameter PPO run	<code>results/best_ppo_real_ohlcvc_2868</code>	Valid
Real OHLCV comparison	<code>results/real_ohlcvc_compare_2854</code>	Invalidated
Optuna search	<code>results/optuna_search_2858</code>	Invalidated

Run 2854 was invalidated because an earlier training entrypoint regenerated synthetic data into real-data paths. Run 2858 was invalidated because the pre-fix sampler repeated parameters across repeated algorithm trials.

7 Synthetic Pilot Results

The latest valid split-aware synthetic comparison is `algo_compare_2853`. The synthetic result confirms that the pipeline can run end-to-end, but the synthetic generator creates simplified market structure and should not be used for market conclusions.

Table 3: Selected synthetic validation results.

Policy	Cum. return	Ann. return	Sharpe	Max drawdown
buy_hold_KBANK.BK	0.580296	0.904531	3.218974	-0.143165
ma_crossover	0.257937	0.381344	1.880703	-0.091600
equal_weight	0.152812	0.221644	1.397638	-0.095300
a2c	0.167278	0.243282	1.372647	-0.103622
mean_variance	0.122426	0.176558	0.860456	-0.109790
ppo	-0.012888	-0.018096	-0.108208	-0.100581

8 Real OHLCV Validation Results

The latest valid real-OHLCV comparison is `real_ohlcv_compare_2855`. On this validation split, the best policy was buy-and-hold PTT.BK. PPO and A2C underperformed equal weight and the strongest single-stock buy-and-hold baselines. PPO had a smaller maximum drawdown than A2C and equal weight in the comparison run.

Table 4: Selected real OHLCV validation results.

Policy	Cum. return	Ann. return	Sharpe	Max drawdown
buy_hold_PTT.BK	0.043523	0.067653	0.405480	-0.103448
buy_hold_ADVANC.BK	0.009999	0.015405	0.097452	-0.098919
equal_weight	-0.030573	-0.046591	-0.416554	-0.103240
a2c	-0.038061	-0.057884	-0.434028	-0.120042
ppo	-0.035900	-0.054629	-0.439532	-0.066219
mean_variance	-0.099682	-0.149008	-1.097376	-0.166737

9 Hyperparameter Search Results

The fixed five-task Optuna search is `optuna_search_fixed_2863`. The best short search trial was PPO trial 4; however, a longer 20,000-timestep run using those parameters produced a cumulative return of -0.030269, annualized return of -0.046132, Sharpe ratio of -0.403028, and maximum drawdown of -0.099395.

The tuned PPO parameters are not yet robust. The short trial improved validation Sharpe, but the longer run regressed. This indicates that the next work should emphasize broader data, walk-forward validation, and larger searches rather than only increasing training time.

10 Walk-Forward Extension

The project now includes configurable walk-forward window generation. The starter real-OHLCV config uses 504 training dates, 126 validation dates, 126 test dates, and a 126-date step size. This produces

Table 5: Fixed Optuna search ranking.

Trial	Algorithm	Sharpe	Cum. return	Max drawdown	Learning rate
4	PPO	0.279954	0.020408	-0.068810	0.000389
0	PPO	-0.305640	-0.002649	-0.008809	0.000017
1	A2C	-0.519583	-0.042686	-0.127149	0.001210
2	PPO	-1.079920	-0.088712	-0.115319	0.002160
3	A2C	-1.708120	-0.003138	-0.004297	0.000029

rolling windows suitable for future one-job-per-window SLURM runs and out-of-sample aggregation.

The first walk-forward pilot is `results/walk_forward_pilot_2869`. It ran two 1,000-timestep PPO jobs through a SLURM array, one task for each generated window. Both error logs were empty. Aggregation wrote combined metrics, a policy summary, and test equity curves.

Table 6: Mean test-window results from the walk-forward pilot.

Policy	Windows	Mean cum. return	Mean Sharpe	Mean max drawdown
<code>buy_hold_ADVANC.BK</code>	2	0.083927	1.467928	-0.064073
<code>buy_hold_KBANK.BK</code>	2	0.064203	1.157021	-0.096110
<code>buy_hold_PTT.BK</code>	2	0.040690	0.672495	-0.081828
<code>equal_weight</code>	2	0.021133	0.633491	-0.068208
<code>ppo</code>	2	0.004260	0.057644	-0.026071

11 Ablation Pilot

The first real-OHLCV ablation pilot is `results/ablation_pilot_2871`. It ran three short 1,000-timestep PPO ablations through a SLURM array: returns only, technical indicators, and technical indicators plus available context placeholders. Aggregation wrote metrics, best-policy, PPO-summary, and bar-chart artifacts.

Table 7: PPO feature ablation pilot.

Feature group	Cum. return	Ann. return	Sharpe	Max drawdown
<code>returns</code>	-0.010482	-0.016061	-0.453371	-0.040127
<code>technical+context</code>	-0.055886	-0.084574	-0.823513	-0.099160
<code>technical</code>	-0.078913	-0.118657	-1.611158	-0.109727

The best policy in each pilot ablation comparison remained `buy_hold_PTT.BK`, with validation Sharpe 0.405480. The ablation machinery is now verified, but stronger claims should wait for broader sector, official macro, sentiment, and fundamentals features.

12 Regime and Stress Tests

The first real-OHLCV regime/stress test is `results/regime_tests_real_ohlcv_20260531`. It evaluates baselines plus the saved PPO/A2C models from `results/real_ohlcv_compare_2855` across named slices and two auto-selected 63-trading-date high-volatility windows.

The acute 2020 COVID crash is explicitly recorded as unavailable because the current starter OHLCV table begins on 2021-01-04. The available COVID-related slice is therefore `covid_recovery_2021`.

Table 8: Best policy by evaluated stress regime.

Regime	Policy	Cum. return	Sharpe	Max drawdown
<code>covid_recovery_2021</code>	<code>buy_hold_ADVANC.BK</code>	0.377423	2.375753	-0.061453
<code>high_volatility_1</code>	<code>buy_hold_ADVANC.BK</code>	-0.016542	-0.735410	-0.055866
<code>high_volatility_2</code>	<code>a2c</code>	0.093690	7.220885	-0.027634
<code>inflation_hike_drawdown_2022</code>	<code>buy_hold_AOT.BK</code>	0.160483	1.366575	-0.047458
<code>test_period</code>	<code>buy_hold_ADVANC.BK</code>	0.483925	3.681822	-0.074324
<code>validation_period</code>	<code>buy_hold_PTT.BK</code>	0.043523	0.405480	-0.103448

13 SET Market-Context Features

The first verified market-context extension uses Yahoo Finance symbol `^SET.BK` for the SET index. The ingestion output has 965 rows from 2021-01-04 to 2024-12-30 and no missing values. The merged feature table has 4,840 stock rows and 26 columns.

The merged table adds SET returns, SET 20-day realized volatility, and a SET moving-average ratio. It also replaces the placeholder `market_return_1d` with the SET index daily return. The first short validation run is `results/real_ohlc_market_pilot_20260531`.

Table 9: Short validation run with SET market-context features.

Policy	Cum. return	Ann. return	Sharpe	Max drawdown
<code>buy_hold_PTT.BK</code>	0.043523	0.067653	0.405480	-0.103448
<code>buy_hold_ADVANC.BK</code>	0.009999	0.015405	0.097452	-0.098919
<code>equal_weight</code>	-0.030573	-0.046591	-0.416554	-0.103240
<code>ppo</code>	-0.054013	-0.081783	-0.952524	-0.104711

This verifies the real SET-index feature path end to end. It does not yet satisfy a broad sector-index requirement because the Yahoo SET50/SET100 candidates probed from HPC did not return usable rows; a more reliable sector-index source is still needed.

14 Sector Context Features

Because Yahoo SET50/SET100 and sector-index candidates were not reliable through the HPC yfinance check, the first sector extension uses a static pilot mapping from SET public listed-company and index constituent classifications. The mapping file is `data/reference/sector_mapping_thai_pilot.csv` and covers the five starter tickers: `ADVANC.BK`, `AOT.BK`, `CPALL.BK`, `KBANK.BK`, and `PTT.BK`.

The sector feature builder writes a 4,840-row, 38-column feature table. The sector data-quality report shows complete panel coverage and zero missing values. Added numeric training columns are sector equal-weight return, sector-relative return, sector peer count, and sector one-hot flags.

The sector-context PPO pilot improves over the SET-market-context PPO pilot on cumulative return, Sharpe, and max drawdown, but it still does not beat the PTT buy-and-hold baseline.

Table 10: Short validation run with SET and sector-context features.

Policy	Cum. return	Ann. return	Sharpe	Max drawdown
buy_hold_PTT.BK	0.043523	0.067653	0.405480	-0.103448
buy_hold_ADVANC.BK	0.009999	0.015405	0.097452	-0.098919
equal_weight	-0.030573	-0.046591	-0.416554	-0.103240
ppo	-0.031097	-0.047383	-0.626349	-0.068484

15 Macro Proxy Features

The first macro extension uses daily Yahoo Finance proxy series rather than official release-date macroeconomic data. The raw macro proxy file has 5,059 rows across USD/THB, Brent crude, WTI crude, gold, and the U.S. 10-year yield. The merge uses forward fill from each proxy’s latest available observation to each Thai stock trading date, so no future macro proxy observations are used.

The macro feature table has 4,840 stock rows and 68 columns. The data-quality report shows complete panel coverage and zero missing values. The first short validation run is the 2026-05-31 macro-proxy pilot.

Table 11: Short validation run with SET, sector, and macro-proxy features.

Policy	Cum. return	Ann. return	Sharpe	Max drawdown
buy_hold_PTT.BK	0.043523	0.067653	0.405480	-0.103448
buy_hold_ADVANC.BK	0.009999	0.015405	0.097452	-0.098919
ppo	-0.022360	-0.034152	-0.344318	-0.085930
equal_weight	-0.030573	-0.046591	-0.416554	-0.103240

The macro-proxy PPO pilot improves over the sector-context PPO pilot and beats equal weight on this short validation run. It still does not beat the PTT or ADVANC buy-and-hold baselines, so it remains a pipeline validation result rather than a model-performance claim.

16 Fundamentals Features

The first fundamentals extension uses Yahoo Finance annual and quarterly financial statements. Quarterly statements are preferred when both annual and quarterly rows exist for the same ticker, period, and metric; annual statements provide older fallback coverage. Features are available only after a 60-day reporting lag.

The raw fundamentals file has 8,701 statement rows across five tickers and 13 period ends. The merged feature table has 4,840 stock rows and 82 columns. Numeric fundamental features are zero-filled before the first lagged report; period markers are missing for 27.7% of rows before coverage begins.

Table 12: Short validation run with fundamentals features.

Policy	Cum. return	Ann. return	Sharpe	Max drawdown
buy_hold_PTT.BK	0.043523	0.067653	0.405480	-0.103448
buy_hold_ADVANC.BK	0.009999	0.015405	0.097452	-0.098919
equal_weight	-0.030573	-0.046591	-0.416554	-0.103240
ppo	-0.025996	-0.039666	-0.416967	-0.067777

The fundamentals PPO pilot has lower max drawdown than equal weight and nearly identical Sharpe, but it still does not beat the PTT or ADVANC buy-and-hold baselines. It also underperforms the macro-

proxy-only pilot, so these fundamentals should be treated as a pipeline validation source until a broader licensed history is available.

17 Official Macro Scaffold

The official macro extension now has a Bank of Thailand source probe and release-date-aware feature merge:

- Raw BOT probe: `data/raw/bot_official_macro.csv`
- Merged feature table: `data/processed/features_real_ohlcvt_official_macro.parquet`
- Data-quality report: `reports/data_quality_official_macro/real_ohlcvt_data_quality.md`

The source probe targets the BOTWEBSTAT EC_EI_002.S2 Leading Economic Indicator table. Monthly observations are assigned release dates using a last-business-day-of-following-month rule, then merged backward-only to stock trading dates. On the current HPC probe, the public web table returned 54 rows across nine indicators, dated 2025-11-01 to 2026-04-01. The modeling table spans 2021-01-04 to 2024-12-30, so there is no overlap. The merged 4,840-row official macro feature table therefore has BOT macro training columns equal to zero throughout the validation period.

Because of that date mismatch, no official macro model-performance claim is made from this probe. The code path is verified by 40 passing HPC tests, but a historical BOT export/API source or downloaded historical file is still required before running a valid official macro ablation.

18 Sentiment Scaffold

The sentiment extension now has source probing, daily scoring, and feature merging paths:

- Raw latest-news probe: `data/raw/news_yahoo_latest.csv`
- Daily sentiment table: `data/processed/sentiment_daily.parquet`
- Merged feature table: `data/processed/features_real_ohlcvt_sentiment.parquet`
- Data-quality report: `reports/data_quality_sentiment/real_ohlcvt_data_quality.md`

The merge is backward-only by ticker/date with a seven-day maximum news age. This prevents future news from being used by earlier stock rows. On the current HPC probe, Yahoo Finance returned 11 latest-news rows across four starter tickers, dated 2025-06-24 to 2026-03-31. The modeling feature table spans 2021-01-04 to 2024-12-30, so there is no overlap. The merged 4,840-row sentiment feature table therefore has `sentiment_score`, `news_count`, and `sentiment_news_age_days` equal to zero throughout the validation period.

Because of that date mismatch, no sentiment model-performance claim is made from this source. The code path is verified by 38 passing HPC tests, but a licensed or otherwise accessible historical Thai news/disclosure source is still required before running a valid sentiment ablation.

19 Conclusion

The implementation is strong enough for a reproducible pilot: ingestion, feature building, leakage checks, chronological splits, DRL training, baselines, plots, hyperparameter arrays, result aggregation, data-quality reporting, sentiment merge scaffolding, official macro release-date merge scaffolding, and walk-forward window configuration all run through a consistent project structure.

The current real-data results do not show PPO or A2C beating the strongest simple buy-and-hold baseline. That is a useful research outcome because it sets a realistic benchmark and shows that the next work should focus on data coverage, multi-source features, walk-forward validation jobs, and ablations.

20 Remaining Work

- Expand the universe beyond five starter .BK tickers.
- Replace the pilot sector mapping with a broader, licensed sector-index or constituent source.
- Add historical official macro data with release-date alignment.
- Replace Yahoo statement fundamentals with licensed SET/SETSMART fundamentals where possible.
- Add historical real news and sentiment features with coverage inside the modeling window.
- Run walk-forward training/evaluation jobs on BistKA.
- Run ablation experiments for price-only, technical, index, macro, sentiment, and all-feature configs.